PERSOONIA
Volume 19 / 2: 261–263 (Published 8 August 2007)

MASSARIOSPHAERIA FRIDAE, A NEW FRESHWATER ASCOMYCETE

M. SPOOREN

Fahrenheitstraat 12, 5621 BR Eindhoven, The Netherlands

A new species of *Massariosphaeria*, hereby proposed as *M. fridae*, is found on submerged stems of *Alisma plantago-aquatica*.

While collecting fungi, an unknown species of *Massariosphaeria* was found. It differs from all other species in that genus by the combination of 4-celled ascospores and its freshwater habitat (Cai et al., 2003).

The ascomata were studied in tap water (crush mounts).

Massariosphaeria fridae M. Spooren, *spec. nov.* — Figs. 1–3

Ascomata gregaria, erumpentia, ostiolo papillato, (sub)globosa ± 500 μm diam., lignum ramorum rubescentia. Paries externus niger, rubescend, e cellulis cubicis, 5 μm diam., seriatis compositus; paries internus pseudoparenchymaticus, hyalinus rubidus. Pseudoparaphyses numerosae, hyalinae, tantum juvenes rubrae, septatae, ramosae, anastomosantes, 1 μm latae. Asci cylindrici, bitunicati, 8-spori, 125–130 × 9 μm, sporae totam longitudinem occupantes. Ascosporae biseriatae, fusiformes, utrinque rotundatae, conicae, hyalinae, (1–)3 septatae, cellula media magis constricta, crassitunicata, sporae maturae stramineae, granulatae, juvenes nonnumquam tunica mucida investitae 25–28 × 6–7 μm.

Habitat ad caules mortuos submersos *Alismatis plantaginis-aquaticae*.

Typus: L 0537613. Eindhoven, Acht, 23-6-2004.

Ascomata gregarious, erumpent, papillate, ostiolate, sphaerioid, ± 500 μm in diameter, staining substrate red. Peridium membranaceous, outer peridium black, with red tinge, composed of rows of cubical cells 5 μm wide. Inner peridium pseudoparenchymatical, hyaline with red tinge. Pseudoparaphyses numerous, hyaline, red when young,

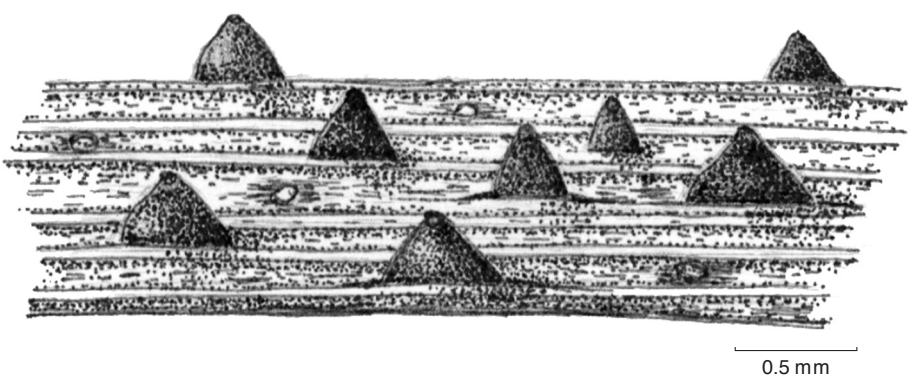
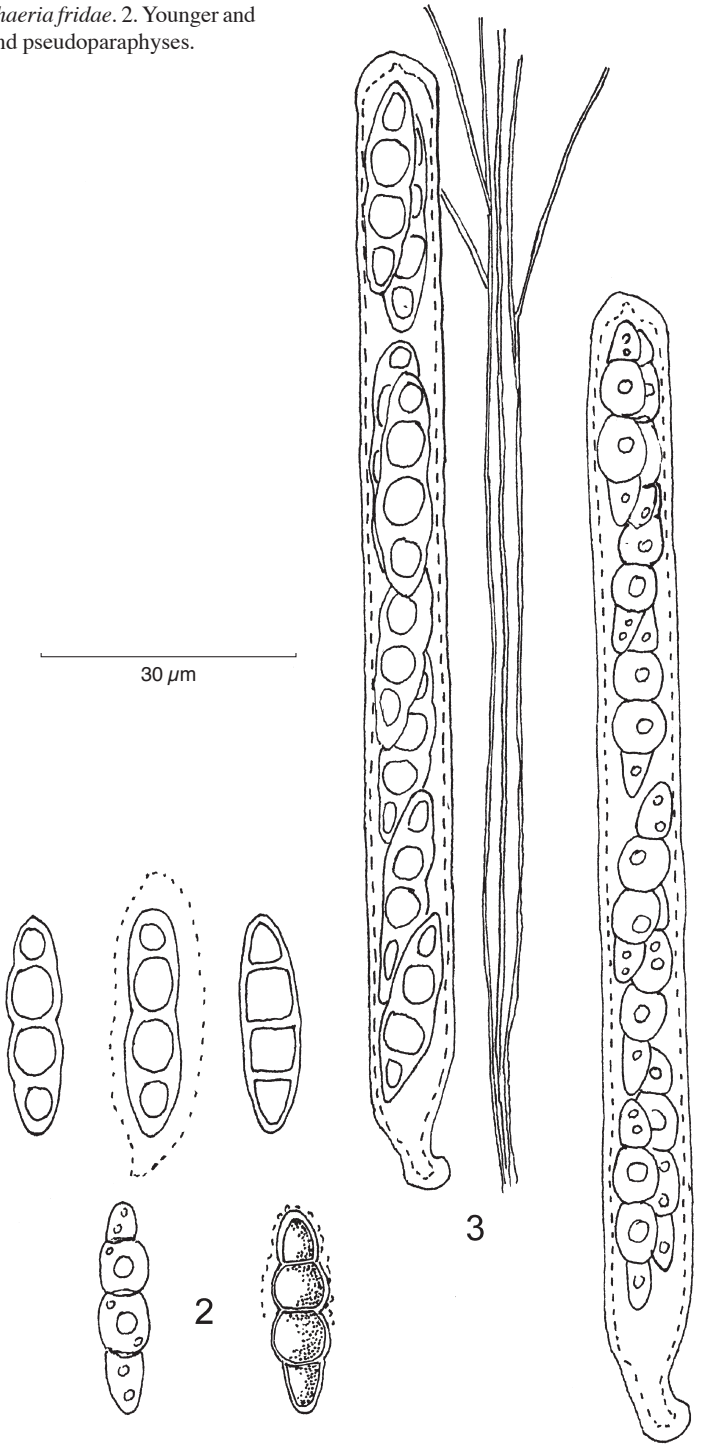


Fig. 1. *Massariosphaeria fridae*. Ascomata.

© 2007 Nationaal Herbarium Nederland, Leiden University branch

Figs. 2, 3. *Massariosphaeria fridae*. 2. Younger and older spores; 3. asci and pseudoparaphyses.



later without colour; septate, branched and anastomosing, 1 μm wide. Asci 125–130 × 9 μm, cylindrical, bitunicate, with 8 ascospores. Spores occupy the whole ascus. Ascospores biseriate, fusiform with rounded conical ends, 3-septate, when young sometimes with 1 or 2 additional septae, strongly constricted at the middle septum; with the two middle-most cells enlarged; when young spores sometimes with a sheath, 25–28 × 6–7 μm; when old straw-coloured and rough.

Habitat — On submerged dead stalks of *Alisma plantago-aquatica*.

Etymology —Thanks to Mrs. Fridi Liebregts-Bax.

Collection examined. Netherlands: Eindhoven, Acht, 23-6-2004, L 0537613.

Massariosphaeria is characterised by the fruit bodies with a sclerotised ostiole and its round to slit-like porus filled with hyaline periphyses. The ascospores have a thick mucilagineous sheath and are straw-coloured and with granulate contents when older. The ascomata have a red pigment and stain the substrate red. Some strains, however, do not stain the substrate, and it is possible that this is also the case with *M. fridae*. They are phragmo-and dictyosporous (Crivelli, 1983; Leuchtmann, 1984), all with more than 5 transverse septa. Leuchtman (1987) describes a 3-septate species from alpine plants.

Massariosphaeria is positioned in the Lophiostomataceae (Kirk et al., 2001).

Massariosphaeria fridae is characterised by a combination of 3-septate ascospores and its freshwater habitat. The species ejaculates under water a slimy mass of hyaline spores and was accompanied by *Phomatospora berkeleyi*, a species known for its facultative aquatic habitat, and a species of *Hymenoscyphus*.

ACKNOWLEDGEMENT

Thanks are due to Mrs. Marianne van Kinsbergen for the drawing and Prof. Dr. W. Gams for the Latin translation of the diagnosis.

REFERENCES

Kirk, P.M., P.F. Cannon, J.C. David & J.A. Stalpers. 2001. Dictionary of the fungi. CABI Surrey.

Cai, L., K.Q. Zhang & K.D. Hyde. 2003. Freshwater ascomycetes. In: C.K.M. Tsui & K.D. Hyde, Freshwater mycology: 275–324. Fungal Diversity Press, Hong Kong.

Crivelli, P.G. 1983. Uber die heterogene Ascomycetengattung Pleospor Rabh.: Vorslag fur eine Aufteilung. Diss. ETH. Zurich nr. 7318.

Leuchtmann, A. 1984. Uber Phaeosphaeria MIYAKE und andere bitunicate Ascomyceten mit mehrfach querseptierten Ascosporen. Sydowia 37: 75–194.

Leuchtmann, A. 1987. Phaeosphaera padellana und Massariosphaeria triseptata, zwei neue bitunicate Ascomyceten aus den Alpen. Mycologia Helvetica 2, 2: 183–191.